

Hassaan Khan, EIT

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EDUCATION

University of Illinois Urbana - Champaign (UIUC) 2025 - present

Doctor of Philosophy in Civil Engineering

- Concentration: Energy-Water-Environment Sustainability (EWES)
- **Dissertation:** Understanding the effects of hydroclimatic extremes across the electricity pipeline and figuring out what ensuring throughput during adverse operating conditions looks like as practical procedures and optimizations relevant to practitioners (*R2O - research to operations*)
- Relevant Coursework: Power System Analysis, Dynamic System Reliability, Probabilistic Risk Analysis, Data Science for the Geosciences, Climate & Social Vulnerability, Hydrocomplexity

University of Illinois Urbana - Champaign (UIUC) 2025

Master of Science in Civil Engineering

- Concentration: Energy-Water-Environment Sustainability (EWES)
- **Thesis:** *Towards connecting occupant behavior to residential water use in a permanent supportive housing context* - [publication](#)

National University of Sciences & Technology (NUST) 2022

Bachelor of Engineering in Civil Engineering

SKILLS

Data Analysis & Languages: Python (pandas, scikit-learn, xarray), MATLAB, Excel (pivot tables)

Applications & Tools: AutoCAD, SAP2000, HEC-HMS, BASEMENT, ArcGIS Pro, EPANET, HAZUS, SWMM, ParaView, Linux Shell

EXPERIENCE

University of Illinois Urbana - Champaign (UIUC) Urbana, IL

Graduate Research Assistant (RA) 05/2025 - present

Working on hydroclimatic extremes and infrastructure systems within the [Stillwell Research Group](#)

Graduate Teaching Assistant (TA) 08/2026 - 12/2026

CEE 433: Water Technology and Policy

Teaching, conducting office hours and grading assignments and exams for a section of 50 students

Graduate Teaching Assistant (TA) 08/2025 - 05/2026

CEE 498: Community Disaster Resilience

Curriculum development and instruction responsibilities for collaborative course on community resilience and disaster planning with stakeholders across CUPHD, ERDC, Champaign County Emergency Management Agency and campus units

Graduate Mentor 06/2025 - 07/2025

AREES (formerly MSI Alliance) Student Summer Research Experience

Graduate Teaching Assistant (TA) 01/2025 - 05/2025

CEE 350: Water Resources Engineering

Teaching, conducting office hours and grading assignments and exams for a section of 53 students

Graduate Research Assistant (RA) 08/2023 - 05/2025

Working on hydroinformatics and residential water within the [Stillwell Research Group](#)

Current Water

National Science Foundation INTERN - [H2NOW Chicago](#)
(H2NOW is a [2026 Fast Company World Changing Idea](#))

Chicago, IL
05/2024 - 08/2024

- Created a predictive model for estimating fecal coliform levels at sites in the Chicago and Calumet rivers
- Assisted with sonde deployment and observation for 2024 season
- Prepared and cleaned previous H2NOW seasons' data for public consumption
- Developed a change-detection script to trigger alerts for turbidity spikes and possible mesh fouling to assist with O&M activities

Educative, Inc.

Editorial (Technical Content Reviewer)

Bellevue, WA
10/2022 - 04/2023

- Edited and reviewed technical course content across 10+ technology areas
- Developed interactive scripts to support technical courses
- Provided input for content curation and development

National University of Sciences & Technology (NUST)

Teaching Assistant (TA)

Islamabad, PK
09/2021 - 01/2022

- [CE-308 Plain and Reinforced Concrete](#)
Assisted with marking quizzes, assignments and assessments; responsible for 70 students
- Helped refine assessment parameters which led to improved test scores for 10+ students

Pakistan Council of Research In Water Resources (PCRWR), MoST

Undergraduate Intern

Islamabad, PK
07/2021 - 09/2021

- Assisted hydrology team with supply, demand and water stress analysis for sector I-10 in consultation with the CDA
- Supported fieldwork with geotagging, classifying and quantifying process parameters for groundwater resources and distribution networks
- Analysed and interpreted results of 7+ designs and scenarios using Hydrus; facilitated R&D operations to explore the efficacy of managed aquifer recharge (MAR) systems in urban settings
- Assessed system reliability and suggested improvements to reduce system downtime by 15% and improve resilience

Spectra Magazine

Editorial Board | [Co-Lead Interview Team](#)

08/2019 - 09/2022

- Led and steered a skilled team of 15+ interviewers and science writers
- Curated popular science content and managed proposals across four teams
- Collaborated with a team of five editors to uphold editorial standards

CERTIFICATIONS

[Adult and Pediatric First Aid/CPR/AED](#)

[FE Exam](#) (TBPELS Certified Engineer-in-Training)

Principles and Practice of Engineering (PE) in Water Resources and Environmental

AWARDS

UIUC Academy for Excellence in Engineering Education (AE3) Teaching Award - Spring 2026

VOLUNTEERING & SERVICE

Engineering Open House

[Special Exhibits Volunteer](#)

04/2024 - present

H2NOW Technical Advisory Committee

Member

05/2024 - present